Eco-evolutionary multitrait biodiversity model

We outline a model to evaluate the effect of trait architecture on biodiversity. Specifically, we implement a trait-based metacommunity model to explore how trait covariation affects the formation of hot and cold spots in biodiversity. We explore two architectures rooted in empirical and theoretical studies. The modular and the magic trait architecture. The modular approach takes into account independent traits. Each trait contributes independently to fitness. The magic trait approach incorporates strong covariance among traits.
Consider many resource and consumer populations characterized each by individuals containing T normally distributed traits. Populations are located in a network of discrete sites connected by migration events and the local population demography is driven by the temporal dependent fitness function accounting for trait architecture following

$$\mathrm{W}(\mathbf{z})_x^t = exp[-\gamma(((\mathbf{z}_x^t - \theta_x)^2)^T \omega^{-1}(\mathbf{z}_x^t - \theta_\mathbf{x})^\mathbf{2})] \,, \tag{1}$$

where $\mathbf{z}_x^t$ is the vector of trait values of phenotype i at time t and site x, θ_x is the multivariate fitness optimum, ω is the covariance matrix (Fig 1) (**??**), and γ determines the interaction sensitivity to deviations from the optimum. If the covariance matrix, ω, is diagonal, representing our modular scenario then there is no correlated stabilizing selection. If the covariance matrix is not diagonal and considering our magic scenario, then there is correlated stabilizing selection with the all traits determining the selection on the other traits.
Biotic optimum and deviations
Phenotypes with dispersal trait value near to this optimum maximize fitness for this trait only for the modular but not for the magic trait scenario.
The abiotic optimum is taken initially as the mean of the distribution and it remains the same during the dynamics (Q2: moving BAD optimum?)
The time dependent fitness function in site x, $\mathrm{W}(\mathbf{z})_x^t$, determines the mother of the offspring. To obtain the phenotypic value for the offspring, $z_o = [z_{ob}, z_{oa}, z_{od}]$, we add to each trait value of the mother the environmental deviation, $\mathbf{e}$, taken from a multivariate uniform distribution with range ± 0.001 in all dimensions (Q3: Take into account additive effects and pleiotropy matrix?). We run our model G generations each following the selection, mating and dispersion scheme to produce a new generation per site. We explore a gradient of dispersal and coevolutionary selection strength, to investigate the formation of hot and cold spots formation in local and regional biodiversity (Fig. 2.)
Interaction matrix The fitness of each individual is calculated with eq 1 – and it determines a weight of the individual in a sampling with replacement function. This means individuals with higher fitness will be producing more offpsring. Phenotypes values of offspring are inherited from the mother with a mutation probability and magnitude sampled from a Poisson and Normal distribution, respectively. Finally, environmental noise is added to the offspring phenotype using a normal distribution with mean 0 and std or var X. Population size of each species in each site for the next generation is determined by a Lotka-Volterra model with community matrix with mean 0 and variance X.